

Name of Student: _____

Scholar No.: _____

INDIAN INSTITUTE OF INFORMATION TECHNOLOGY BHOPAL
Department of Mathematics

Branch: CS (AI)/Section 4

Semester–I

Mini-Test – 5 October 2026

Subject Code: CSE-1001

Max. Marks: 20

Subject Name: Engineering Mathematics – I

Time: 60 Minutes

NOTE:

1. All questions are compulsory.
2. The obtained marks will be divided by 2 and counted as your mini-test marks, which are out of 10.

S. No.	Question	Marks
1 (A)	(a) Give an example of a function $f : \mathbb{R} \rightarrow \mathbb{R}$ which is continuous on $\mathbb{R}$ and differentiable at every point except at each integer. (b) Consider the function $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = \begin{cases} x \sin\left(\frac{1}{x}\right), & x \neq 0, \\ 0, & x = 0. \end{cases}$ Determine, with justification, whether f is differentiable at $x = 0$.	1 + 1
(B)	(a) Let $f : \mathbb{R} \rightarrow \mathbb{R}$ satisfy $f(x + y) = f(x)f(y) \quad \text{for all } x, y \in \mathbb{R}.$ Given that f is differentiable at $x = 0$, prove that f is differentiable at every $x \in \mathbb{R}$. (b) Determine the value of $k \in \mathbb{R}$ for which the function $f(x) = \begin{cases} x^2, & x \leq 1, \\ kx, & x > 1, \end{cases}$ is continuous at $x = 1$. With this value of k, is f differentiable at $x = 1$? Justify.	2 + 1

S. No.	Question	Marks
2 (A)	(a) Give, with justification, one example of a pair of sequences (a_n), (b_n) of real numbers such that (i) both (a_n) and (b_n) diverge, but $(a_n + b_n)$ converges; (ii) both (a_n) and (b_n) diverge, but $(a_n b_n)$ converges to a non-zero limit; (iii) (a_n) converges, (b_n) diverges, and $(a_n b_n)$ converges; (iv) $a_n \rightarrow 0$, $b_n \rightarrow \infty$, and $a_n b_n \rightarrow 5$. (b) Examine the convergence of the sequence $a_n = (1^n + 2^n + 3^n)^{1/n}$, and determine its limit.	2 + 1
(B)	Evaluate $\lim_{n \rightarrow \infty} \left(\frac{1}{n^2 + 1} + \frac{2}{n^2 + 2} + \frac{3}{n^2 + 3} + \cdots + \frac{n}{n^2 + n} \right).$	2
3 (A)	Test the convergence of each of the following series, naming the test used in each case: (i) $\sum_{n=2}^{\infty} \frac{1}{n (\ln n)^2}$ (ii) $\sum_{n=1}^{\infty} (\sqrt{n^2 + 1} - n)$ (iii) $\sum_{n=1}^{\infty} \frac{(-1)^{n+1} n}{n^2 + 1}$ (examine for absolute and conditional convergence)	3
(B)	Discuss the convergence of the series $\sum_{n=1}^{\infty} \frac{(3x - 1)^n}{n 4^n},$ stating the interval of convergence with both endpoints tested separately.	2
4 (A)	Obtain the linear and the quadratic approximation to $f(x) = \sqrt{x}$ about $a = 1$, and use them to estimate $\sqrt{1.1}$. Using the Remainder Estimation Theorem, obtain an upper bound for the error in each case.	3
(B)	Obtain the Maclaurin series of $\cos x$ and state the values of x for which it is valid.	2

ALL THE BEST